RAJALAKSHMI ENGINEERING COLLEGE

Priyadharshini K 251801248-2025-AIDS

P2

Started on	Wednesday, 2 September 2026, 7:01 PM
State	Finished
Completed on	Wednesday, 2 September 2026, 7:06 PM
Time taken	5 mins 6 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[], int l, int h, int x) {
    if (l > h)
        return -1;

    int m = (l + h) / 2;

    if (a[m] == x)
        return a[m];

    if (a[m] > x)
        return floorValue(a, l, m - 1, x);

    int r = floorValue(a, m + 1, h, x);

    return (r == -1) ? a[m] : r;
}

int main() {
    int n, x;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course